Digital Elevation Model (DEM) for secondary mission products of BIOMASS mission

Microwaves and Radar Institute
German Aerospace Center (DLR)

BIOMASS - ESA's 7th Earth Explorer Mission

2005 EE7 Call

2006 BIOMASS Proposal Submission

2007 Phase 0 (6 Mission Proposals)

2009 User Consultation Meeting Lisbon

2010 Phase A (3 Mission Proposals)

2013 User Consultation Meeting Graz

2013 Phase B (BIOMASS Only)

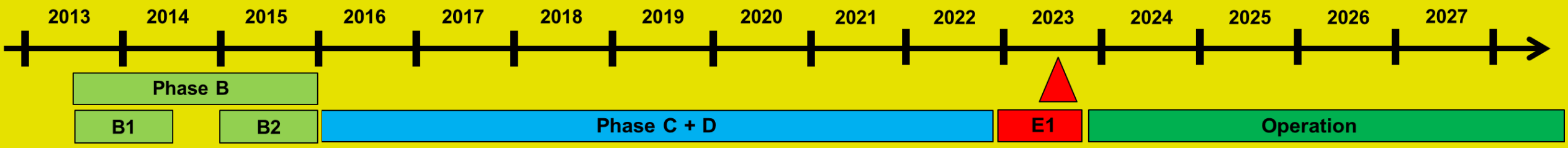
2016 Phase C+D

2023 Launch

2023-2028 Operation

K.Papathanassiou, DLR

on behalf of the BIOMASS Mission Advisory Group

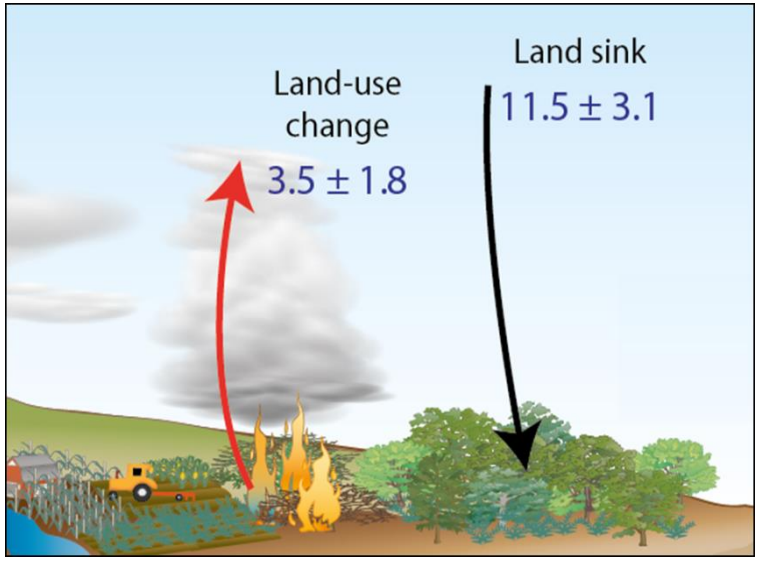


Mission Objectives

1. **Reduce the major uncertainties in terrestrial Carbon fluxes** due to Land Use Change, forest degradation & regrowth.
2. Provide support for International Agreements (UNFCCC & REDD+).

Secondary objectives

1. Sub-surface geology in deserts
2. Glacier and ice sheet velocities
3. Topographic mapping under dense vegetation



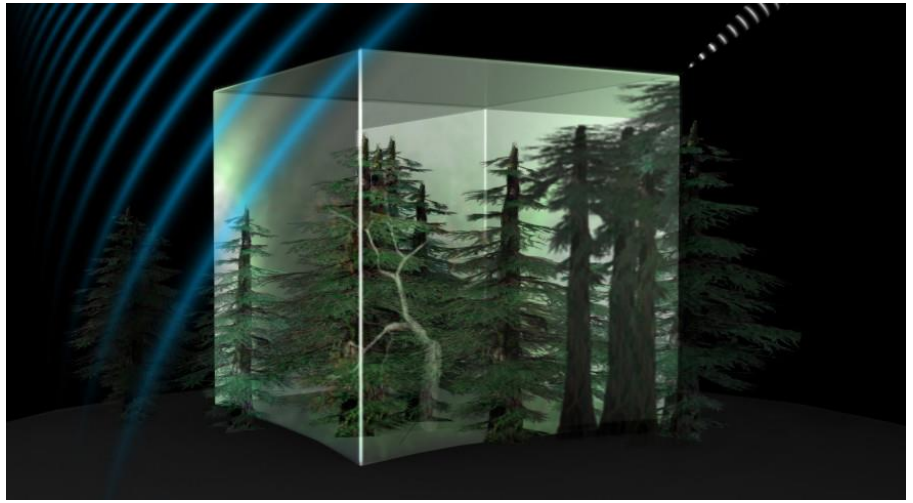
Gigatonnes of Carbon dioxide per year

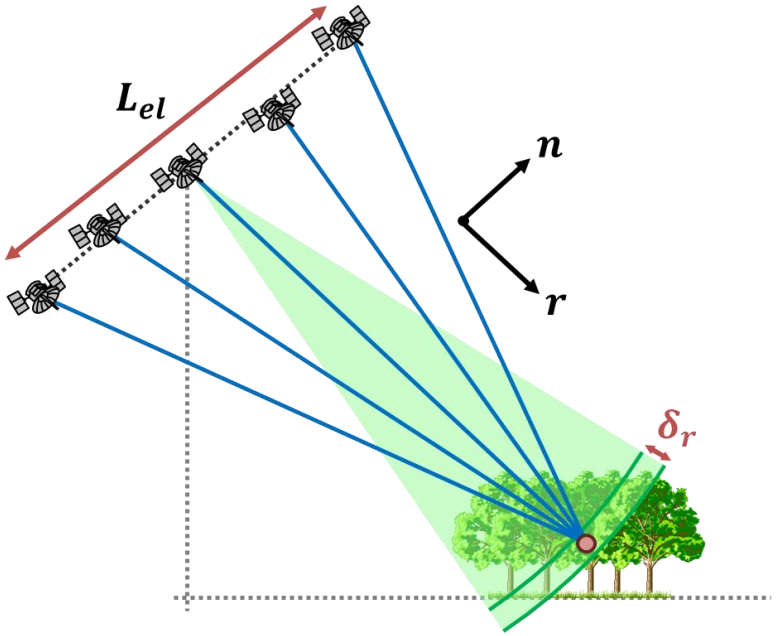
What is needed for this:

1. Estimates of **forest biomass, height and disturbance.**
2. **T**he **crucial information need** (in terms of deforestation and regrowth) is in the **tropics**.
3. **Biomass measurements are needed** where the changes occur and **at the effective scale of change** (4 hectares).
4. Detection of **deforestation at 0.25 ha.**
5. **A biomass accuracy of 20% at 4 hectares**, comparable to ground-based observations.
6. **Repeated measurements** (preferable twice per year) over multiple years to asses deforestation & regrowth.

Motivation for a BIOMASS DEM/DTM

- Generation of digital terrain models (DTM) ⇒ elevation of the ground without vegetation
- At P band, vegetation causes less attenuation ⇒ phase center closer to the ground, strong ground return
- Applications: ecology, forest management, water resource management, mineral exploitation, etc.
- Possibility to exploit DTM for primary objectives (e.g., slope correction)





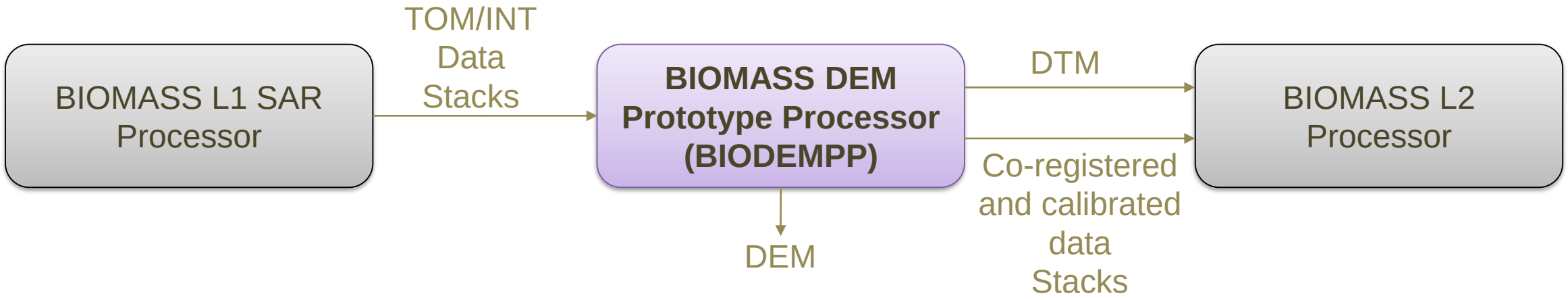
Context

- ESA study: **BIOMASS DEM Product Prototype Processor**
  - March 2019 – June 2021
  - Consortium: DLR (Prime), Aresys, Politecnico Milano

DLR	aresys	POLITECNICO MILANO 1863	esa
Muriel Pinheiro Pau Prats Joel Amao Nida Sakar Gustavo Martin del Campo Becerra Matteo Nannini Rolf Scheiber Alberto Alonso Marc Jäger Nestor Yague-Martinez Kostas Papathanassiou	Simone Mancon Davide Giudici Riccard Piantanida Francesco Banda	Mauro Mariotti d'Alessandro Stefano Tebaldini	Klaus Scipal

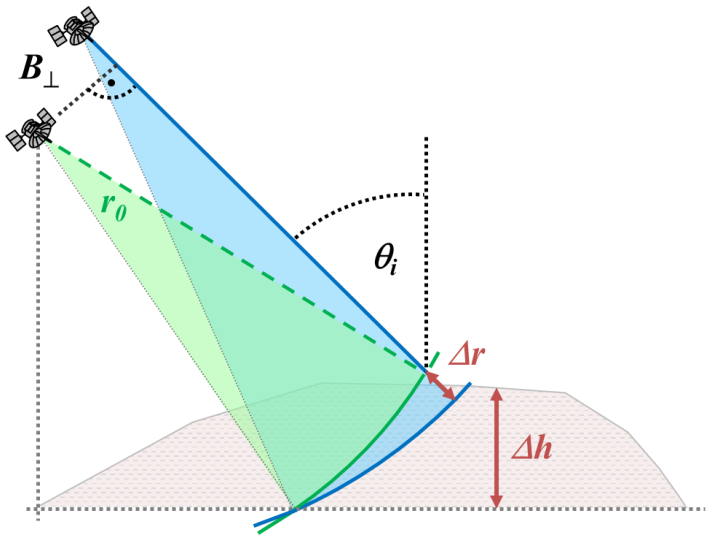
Context

- **Main outputs**:
  - DTM (ground topography)
  - Co-registered and calibrated stacks
  - DEM (phase center height at P band)



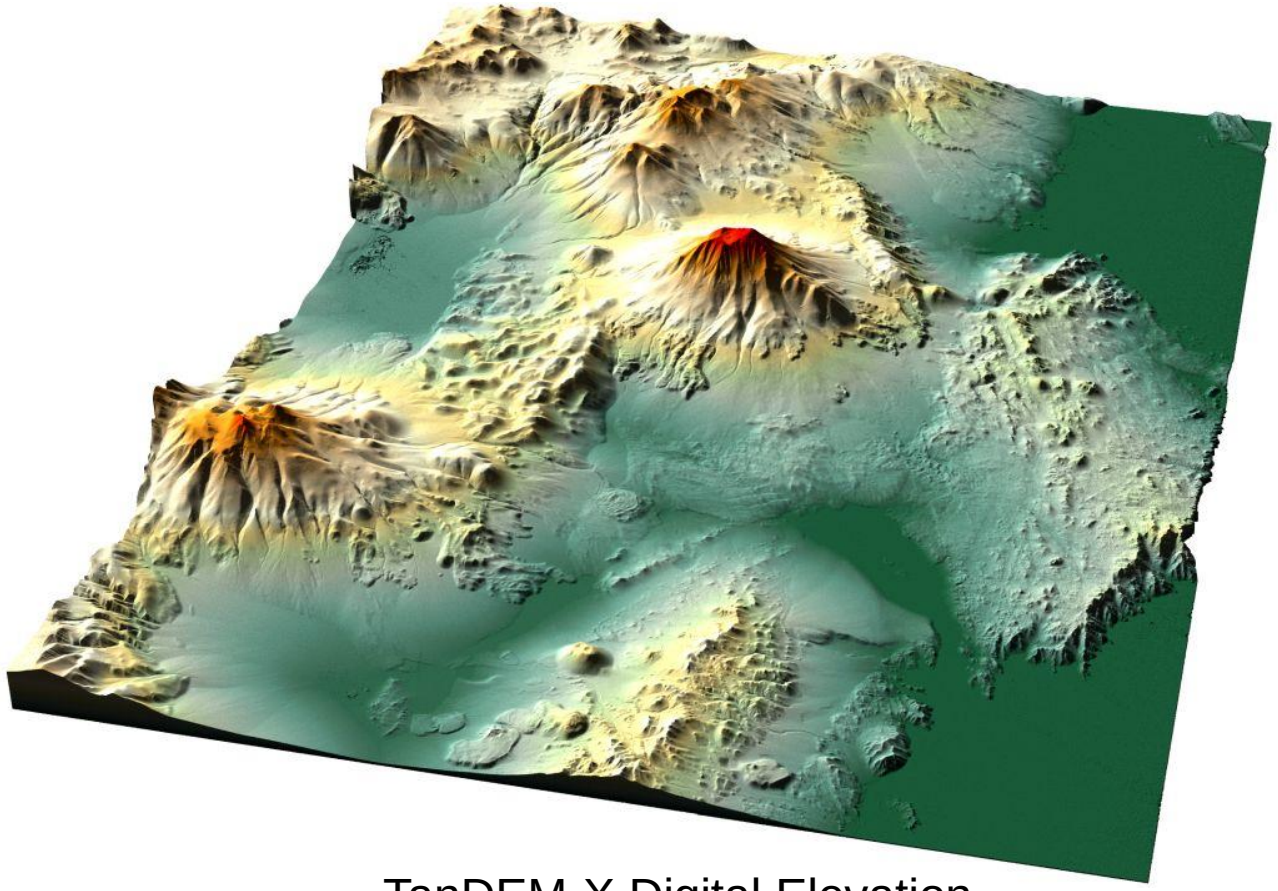
SAR Interferometry and SAR Tomography

SAR Interferometry



Retrieves mean height of resolution cell with high accuracy

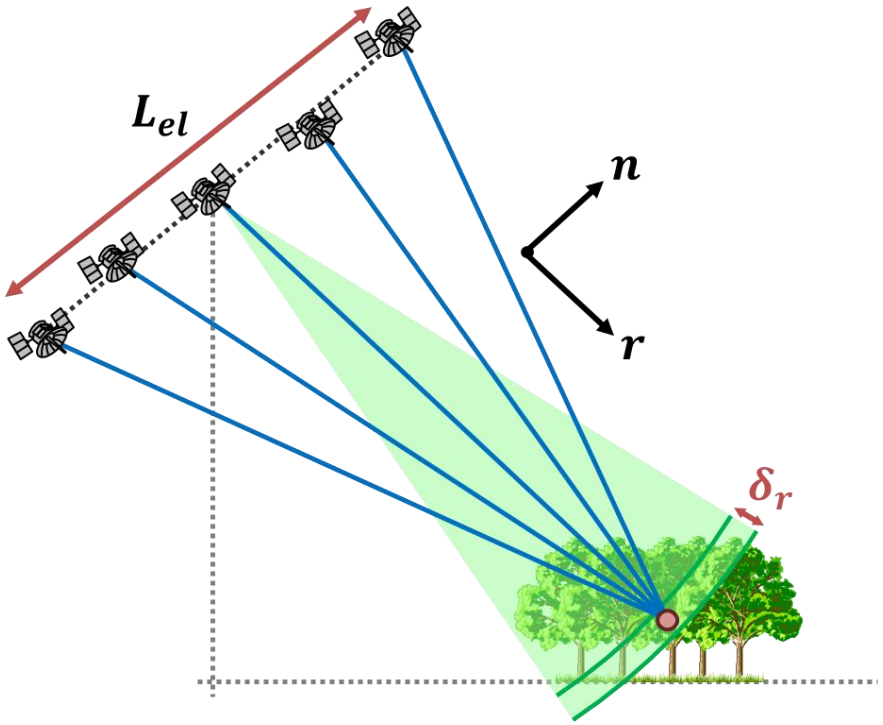
Multi-baseline approaches improve robustness



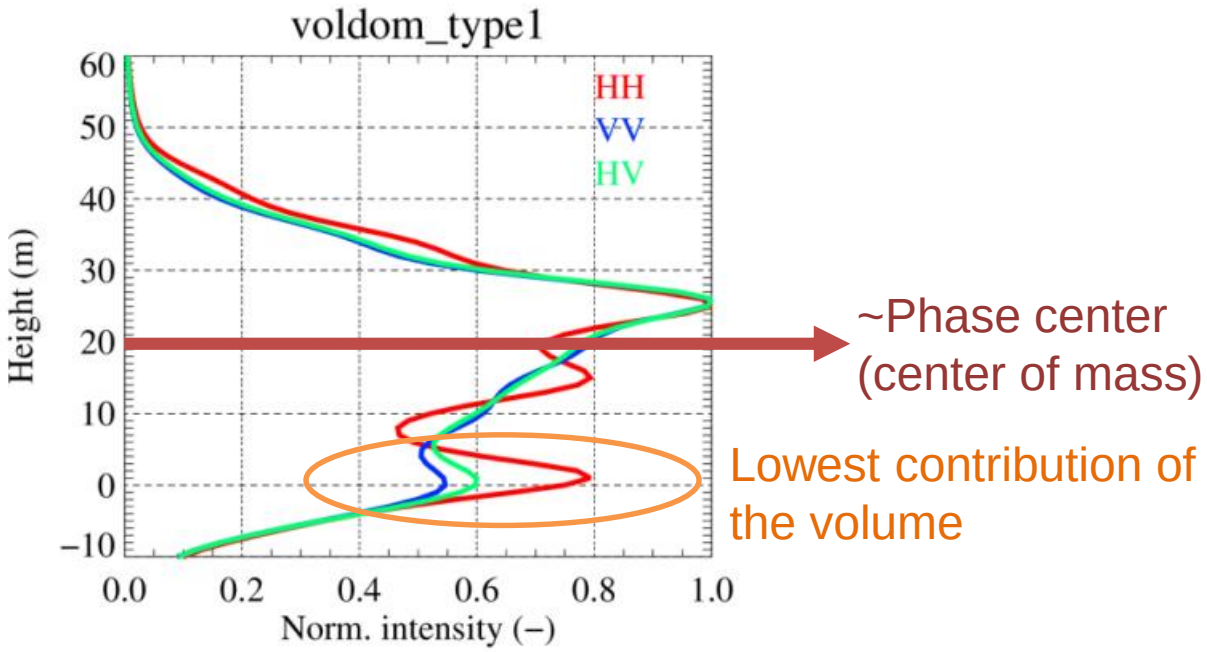
TanDEM-X Digital Elevation Model over the Atacama desert

SAR Interferometry and SAR Tomography

SAR Tomography

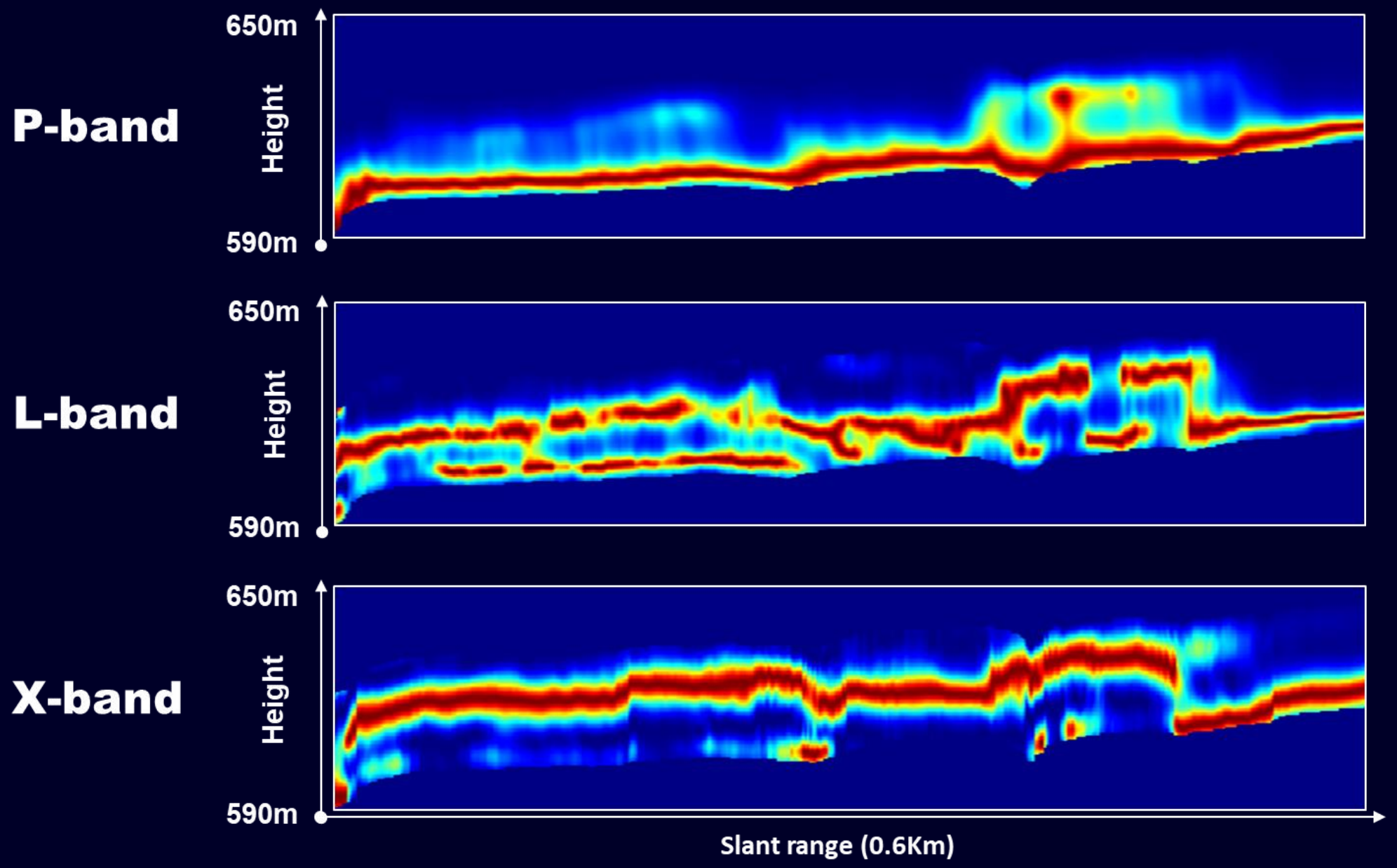


Retrieves vertical reflectivity profile with limited resolution (>23 m for BIOMASS)



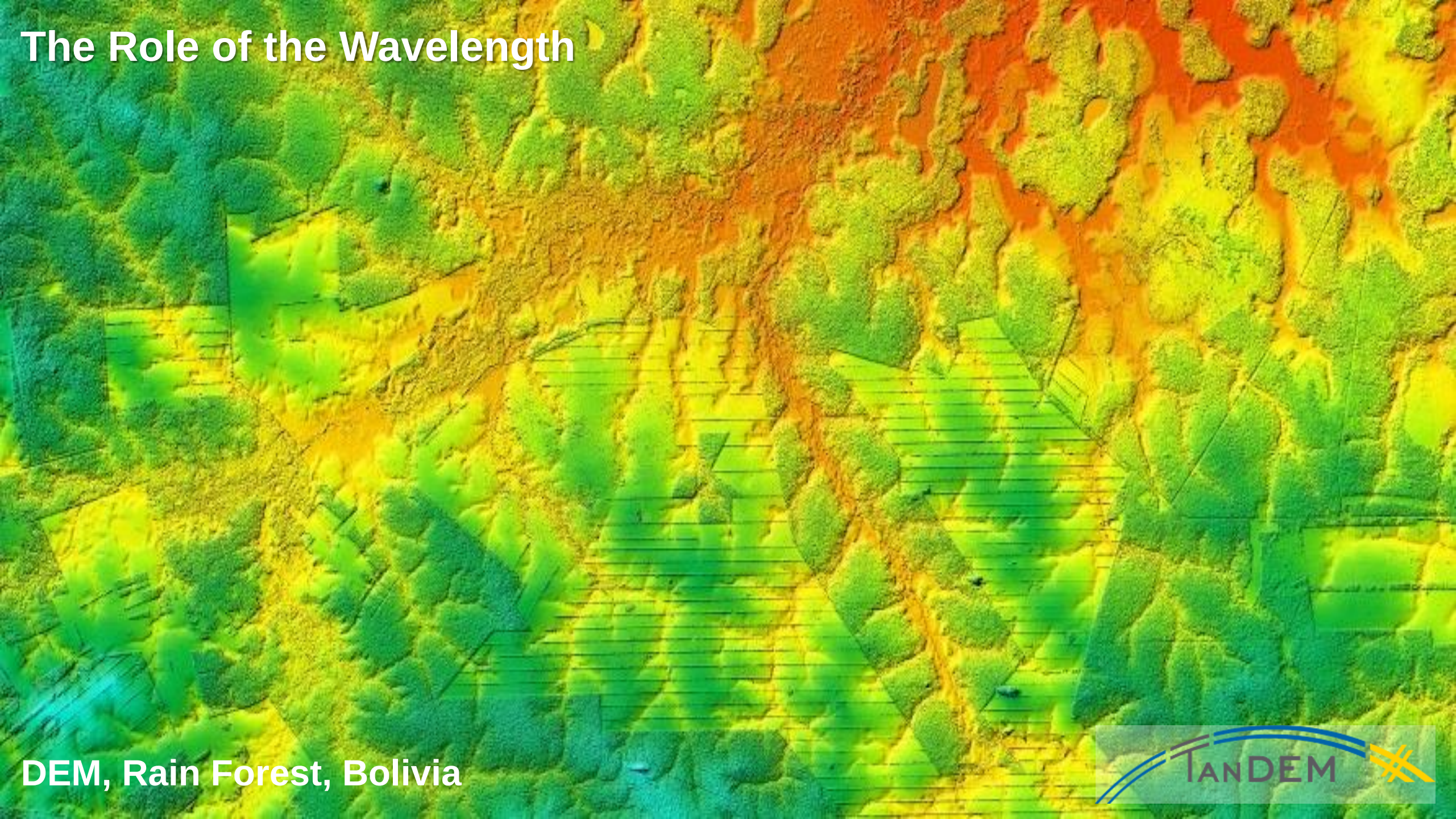
The BIODEMPP combines interferometric and tomographic information to generate the DTM

Traunstein forest (Germany) – HH Tomograms



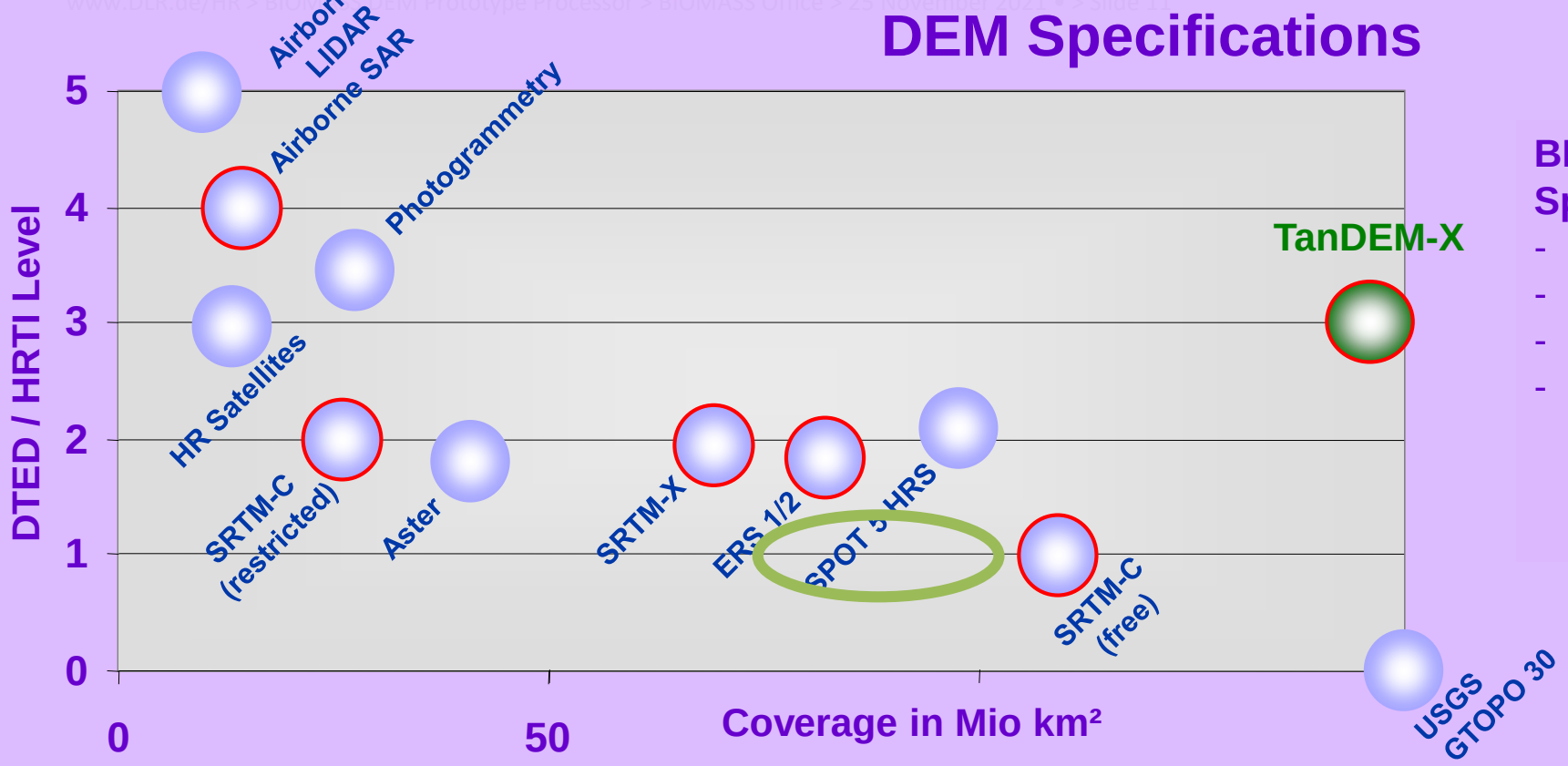
DLR

The Role of the Wavelength



DEM, Rain Forest, Bolivia

DEM Specifications



BIOMASS DEM/DTM Product Specification (TBC):

- Spatial resolution: ~100 m x 100 m
- Relative vertical accuracy: < 12 m
- Absolute vertical accuracy: < 18 m
- Limited coverage due to space object tracking radar (SOTR) restrictions (not covered: Europe, North America, North Pole incl. Greenland)

	Spatial Resolution	Absolute Vertical Accuracy (90%)	Relative Vertical Accuracy (point-to-point in 1°cell, 90%)
DTED-1	90 m x 90 m	< 30 m	< 20 m
DTED-2	30 m x 30 m	< 18 m	< 12 m
HRTI-3	12 m x 12 m	< 10 m	< 2 m
HRTI-4	6 m x 6 m	< 5 m	< 0.8 m

NGA (NIMA) Standards for Digital Elevation Models

BIOMASS Mission Coverage

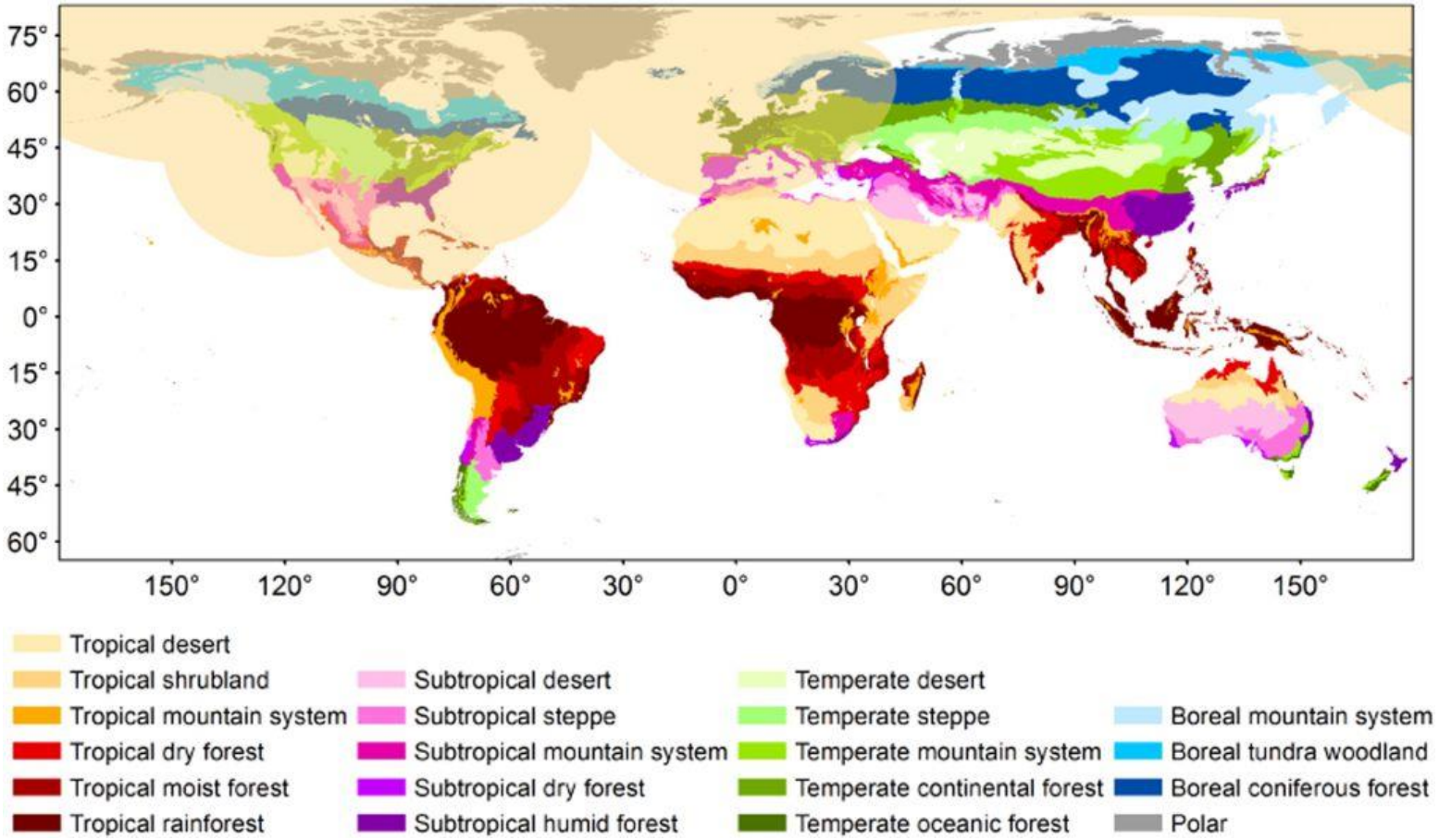
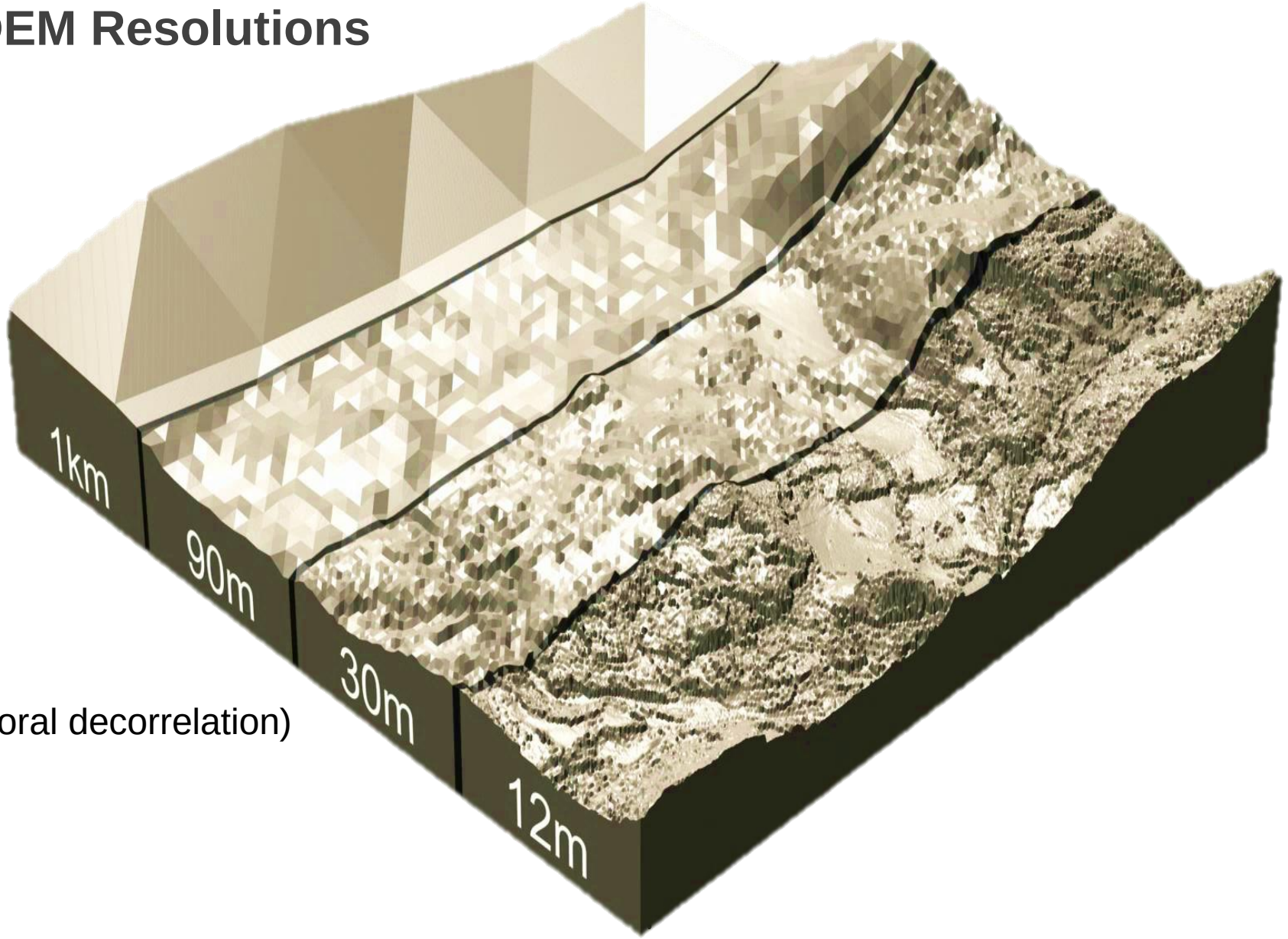


Image Source: Carreiras, J. M., Quegan, S., Le Toan, T., Minh, D. H. T., Saatchi, S. S., Carvalhais, N., Reichstein, M., Scipal, K. (2017). Coverage of high biomass forests by the ESA BIOMASS mission under defense restrictions. Remote Sensing of Environment, 196, 154-162.

Comparison of spatial DEM Resolutions



BIOMASS DEM/DTM challenges:

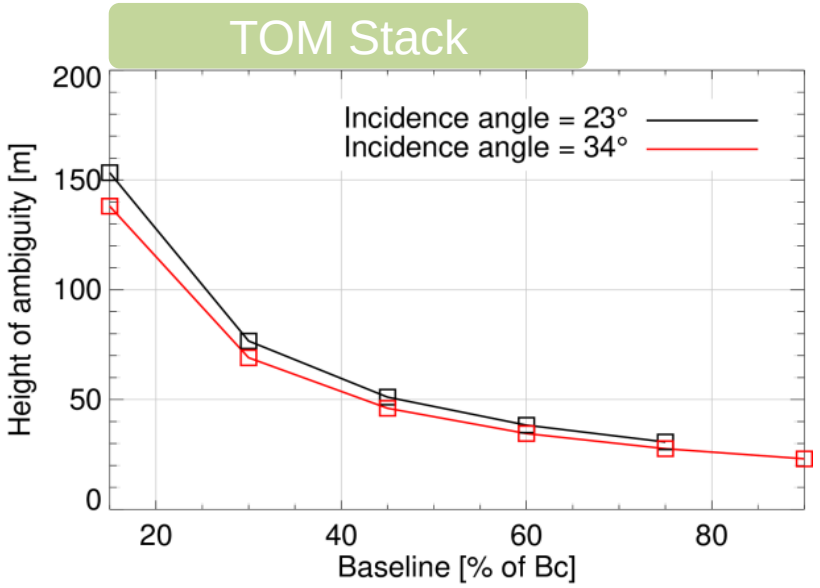
- Repeat-pass acquisitions (temporal decorrelation)
- Ionosphere & troposphere
- Vegetation bias

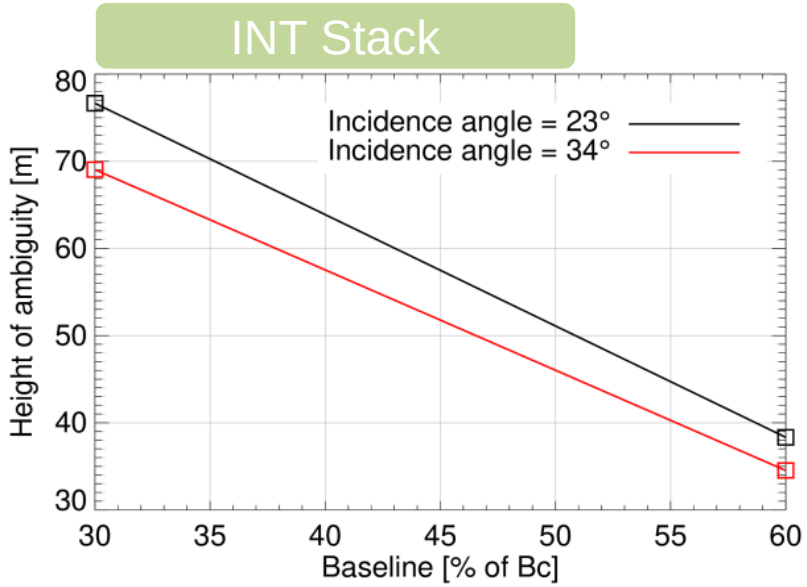
The TOM and INT Phases

- The TOM Phase
  - First year of the mission
  - 7 acquisitions per swath
  - Repeat pass of 3 days
  - Baseline increments of 15% of critical baseline

- The INT Phase
  - Remaining 4 years of mission
  - 3 acquisitions per swath (once every 6 months)
  - Repeat pass of 3 days
  - Baseline increments of 30% of critical baseline

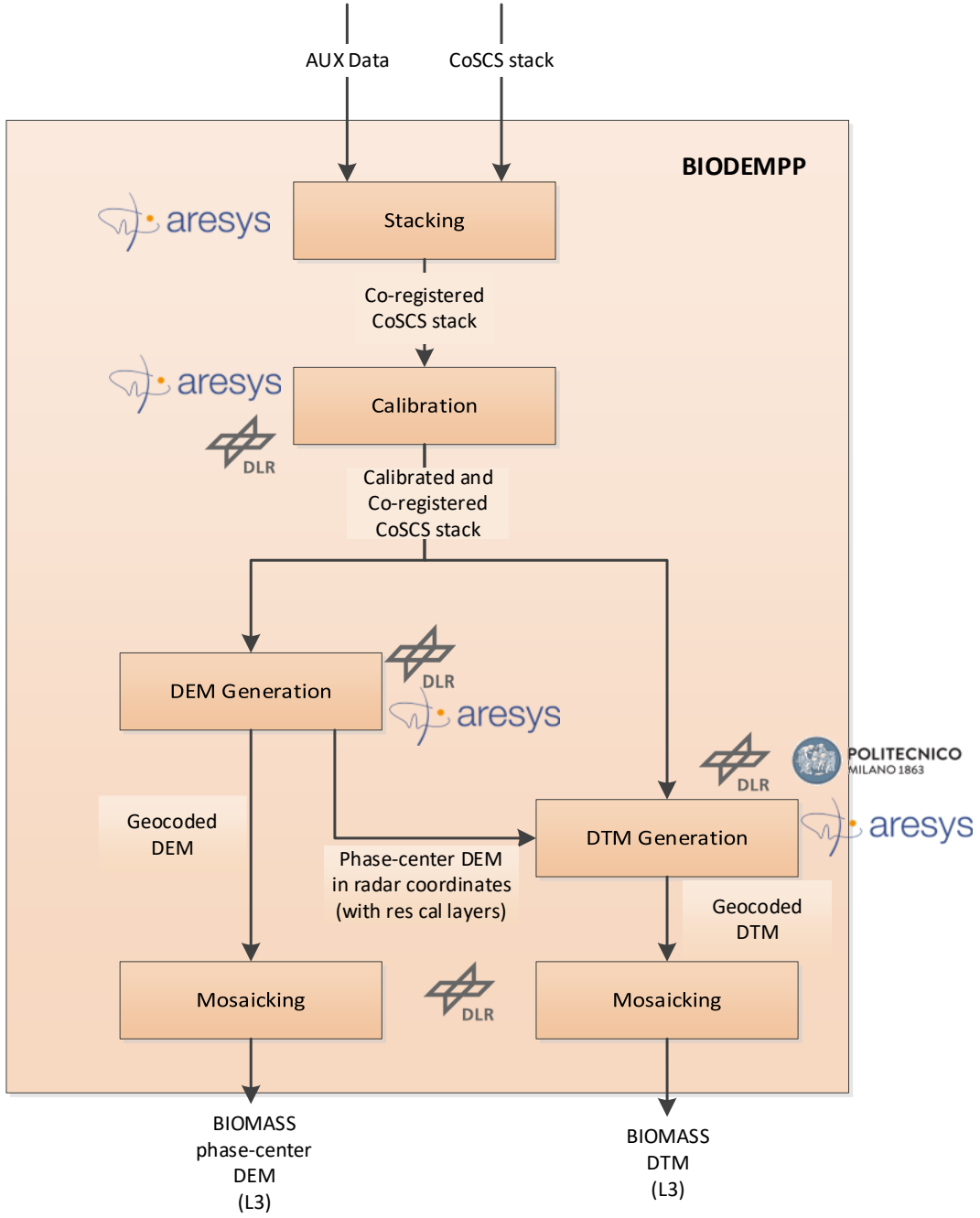
The BIODEMPP can work with data acquired with both the TOM and INT phases, but the results with the TOM phase will be more robust and accurate.





Processing Flow

1. Stacking: alignment of all images to the same geometry
2. Calibration:
   - Correction of differential ionosphere
   - Correction of baseline errors
   - Residual calibration (external reference)
3. DEM generation:
   - Spectral filtering & coherence optimization
   - Multi-baseline height estimation
   - Geocoding
4. DTM generation:
   - SKP to retrieve ground phase
   - Multi-baseline height estimation
   - DTM/DEM fusion and geocoding
5. Mosaicking of swath-based DEMs and DTMs

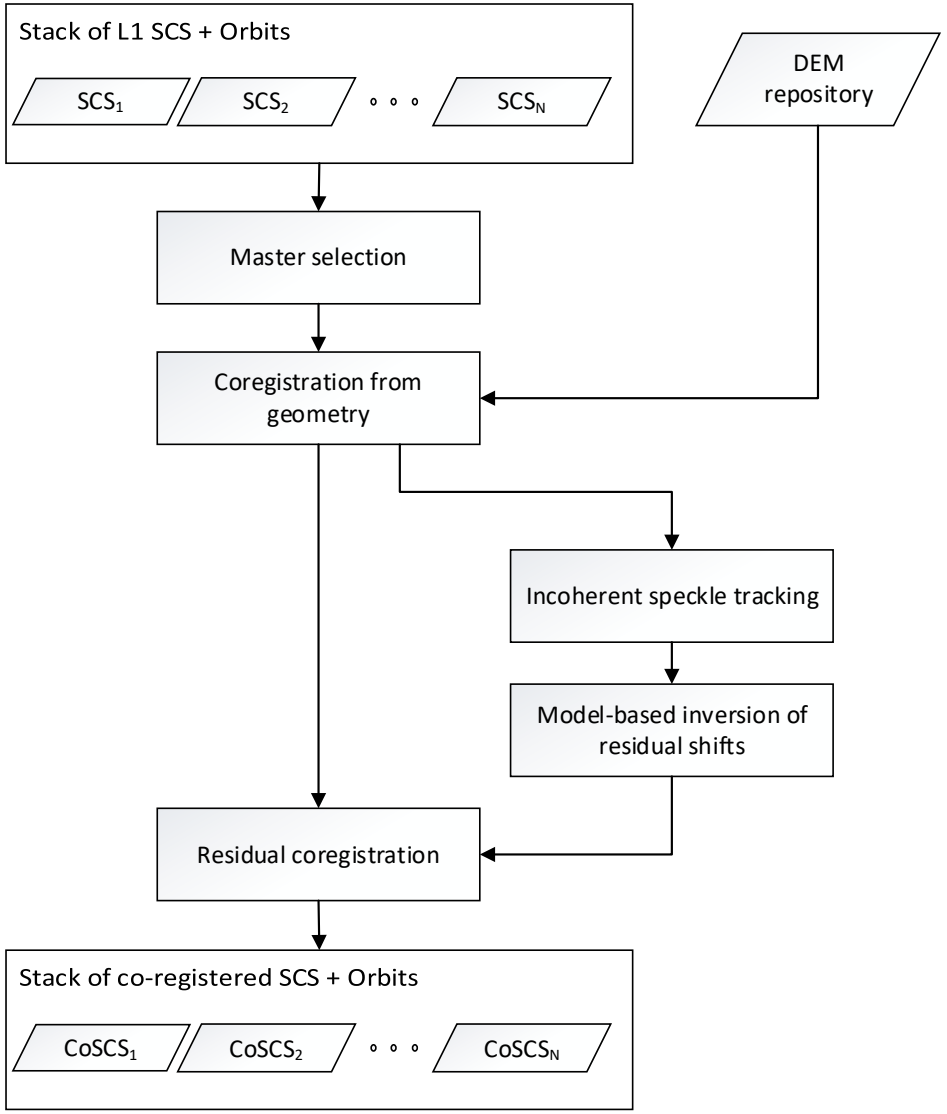


Stacking Module

Co-registration of the interferometric stack

- Co-registration allows having the L1a images sampled on the same slant-range/azimuth grid.

- Due to the expected poor quality of the orbit knowledge and the strong sensitivity to the ionosphere two co-registration steps are needed:
  - Co-registration from geometry (orbit + DEM)
  - Co-registration from data (incoherent speckle tracking)

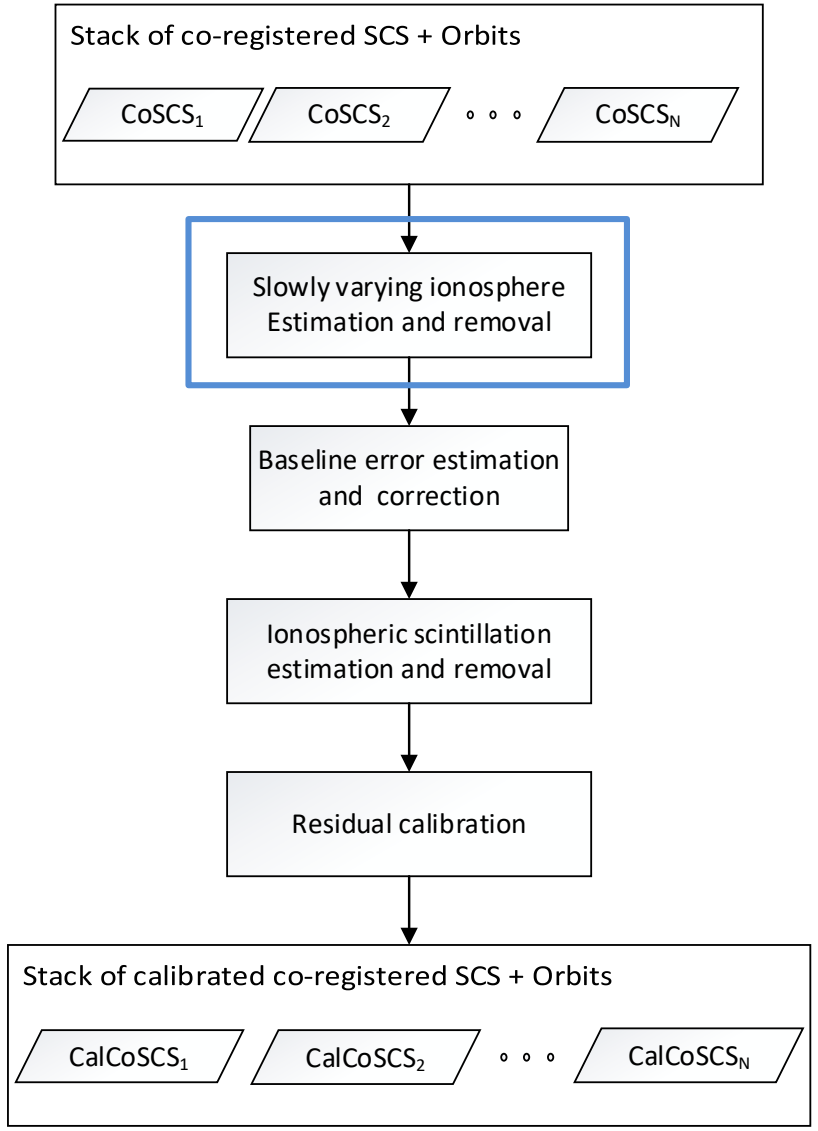


Calibration Module

Calibration of the interferometric stack

- The aim of the calibration step is to mitigate the phase artifacts that can impact on the performances of the multi-baseline height estimation and generate the calibrated stack

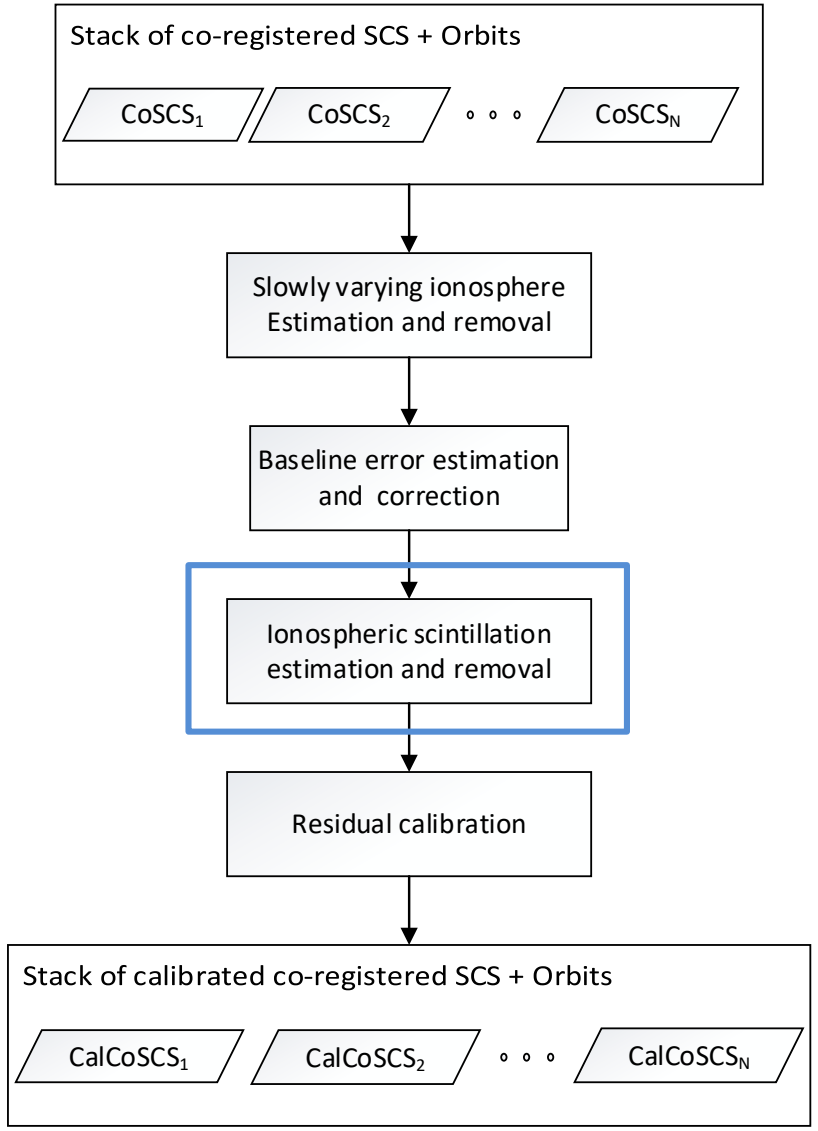
- Background ionosphere estimation with split-bandwidth technique (multibaseline approach)



Calibration of the interferometric stack

- Residual estimation/removal of ionosphere scintillation (w.r.t. to L1 correction)
- Uses multi-squint and multi-baseline information
- For details on the algorithm refer to:

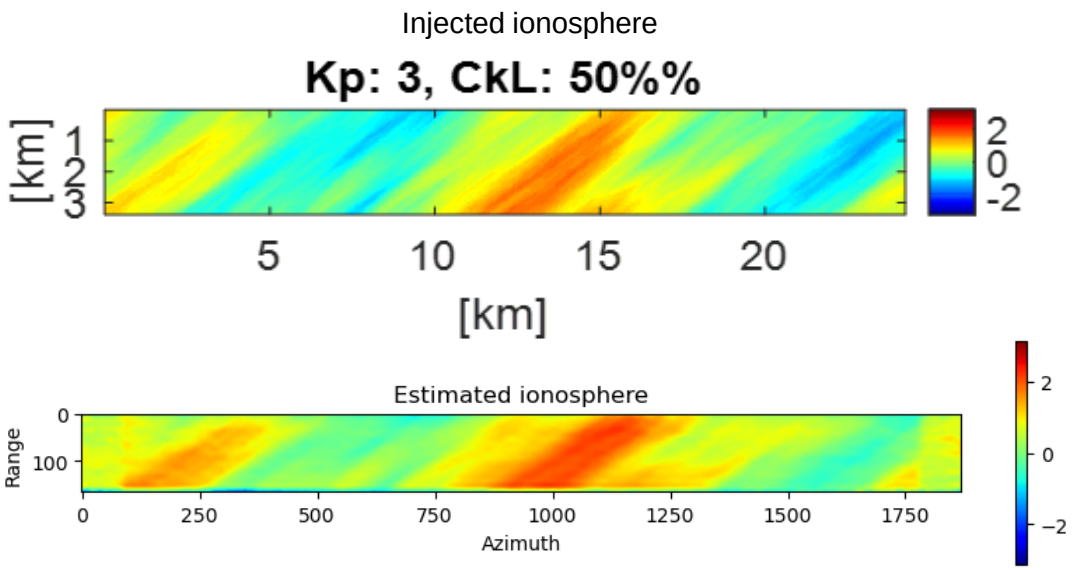
S. Mancon, D. Giudici and S. Tebaldini, "The ionospheric effects mitigation in the BIOMASS mission exploiting multi-squint coherence supported by Faraday rotation," *EUSAR 2018; 12th European Conference on Synthetic Aperture Radar*, 2018, pp. 1-6.

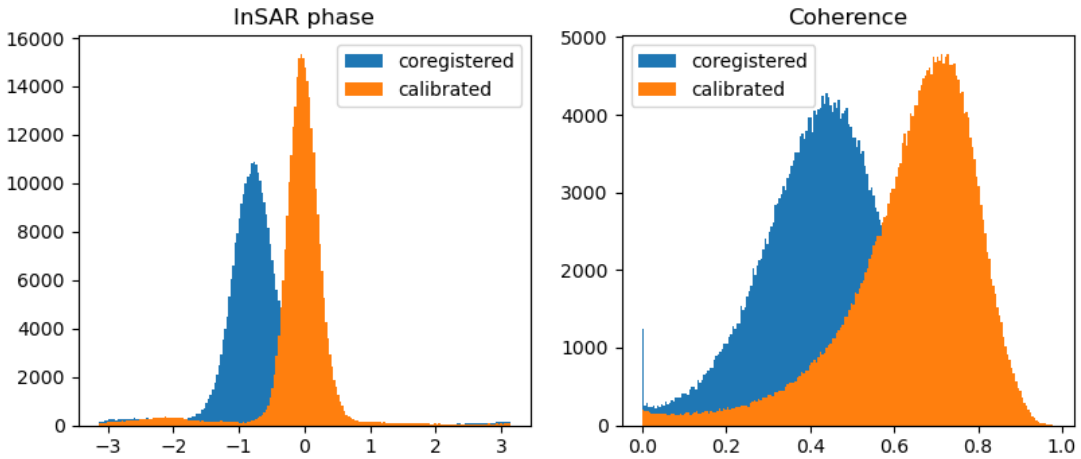


Calibration of the interferometric stack – ionospheric scintillation

Sample result:

- Interferometric pair has been simulated in BIOMASS geometry from airborne data
- Ionospheric scintillation phase screen has been injected in one of them
- The calibration step allows improving both the coherence and the interferometric phase.



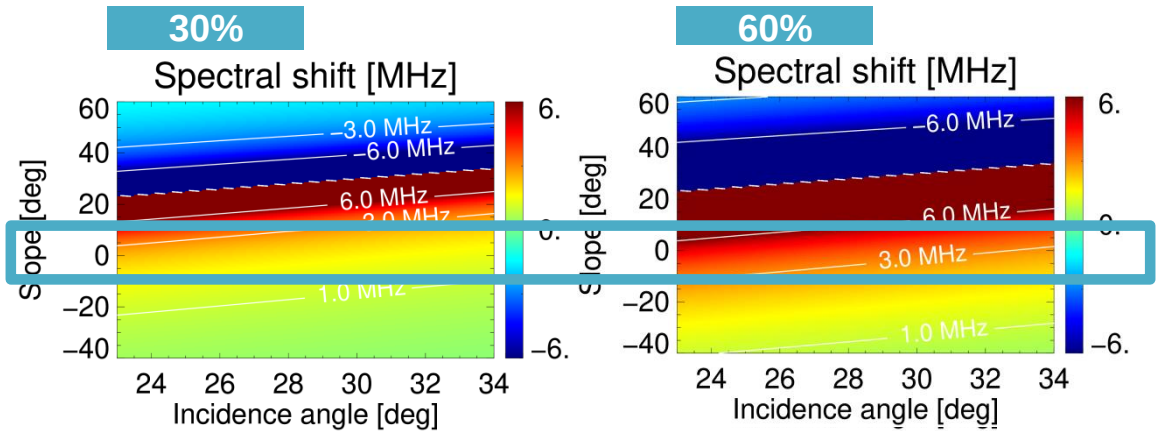


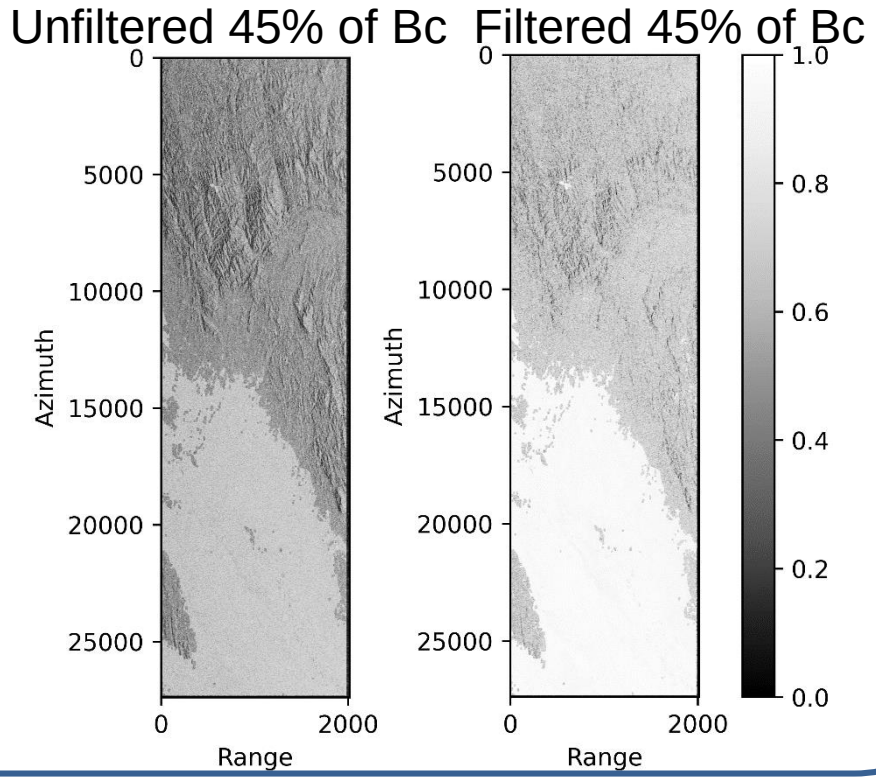
DEM Generation

Decorrelation

- Decorrelation due to poor backscatter, geometry or volume
- Tackled in three steps: spectral filter, coherence phase optimization & multi-baseline phase center height estimation

Spectral filter





Decorrelation

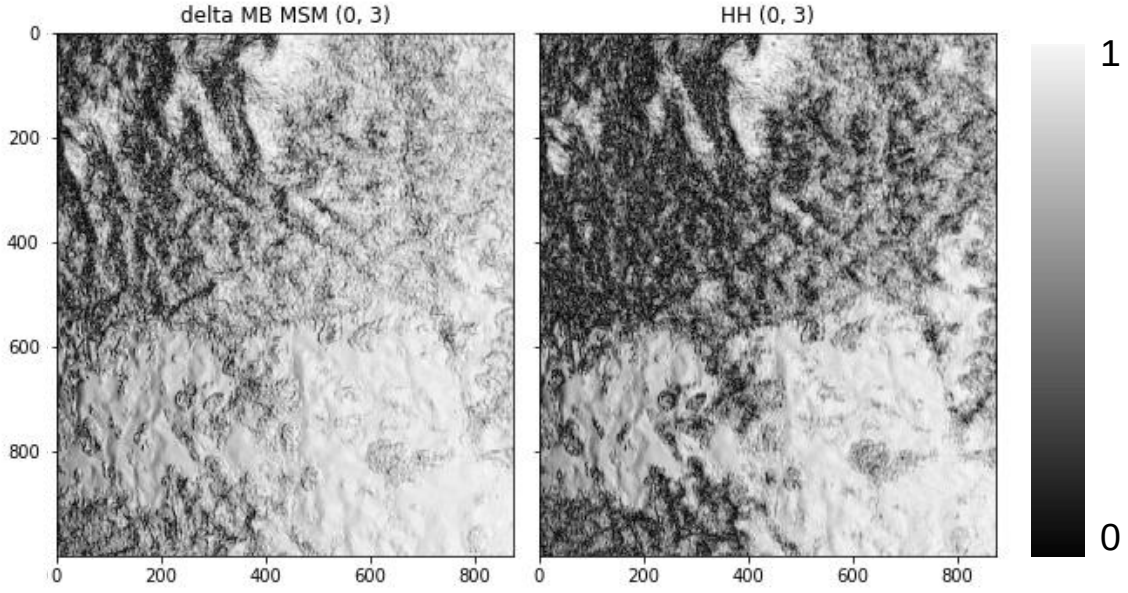
- Decorrelation due to poor backscatter, geometry or volume (specially for large baselines w.r.t. critical)
- Tackled in three steps: spectral filter, coherence phase optimization & multi-baseline phase center height estimation

Coherence phase optimization

$$\delta_\phi = \left| \frac{1}{N} \sum_{i=0}^{N} e^{j\phi_i} \right|$$

where $\phi_i = \arg \gamma_i^{opt}$

[2] M. Neumann, et al, "Multibaseline Polarimetric SAR interferometry coherence optimization,"

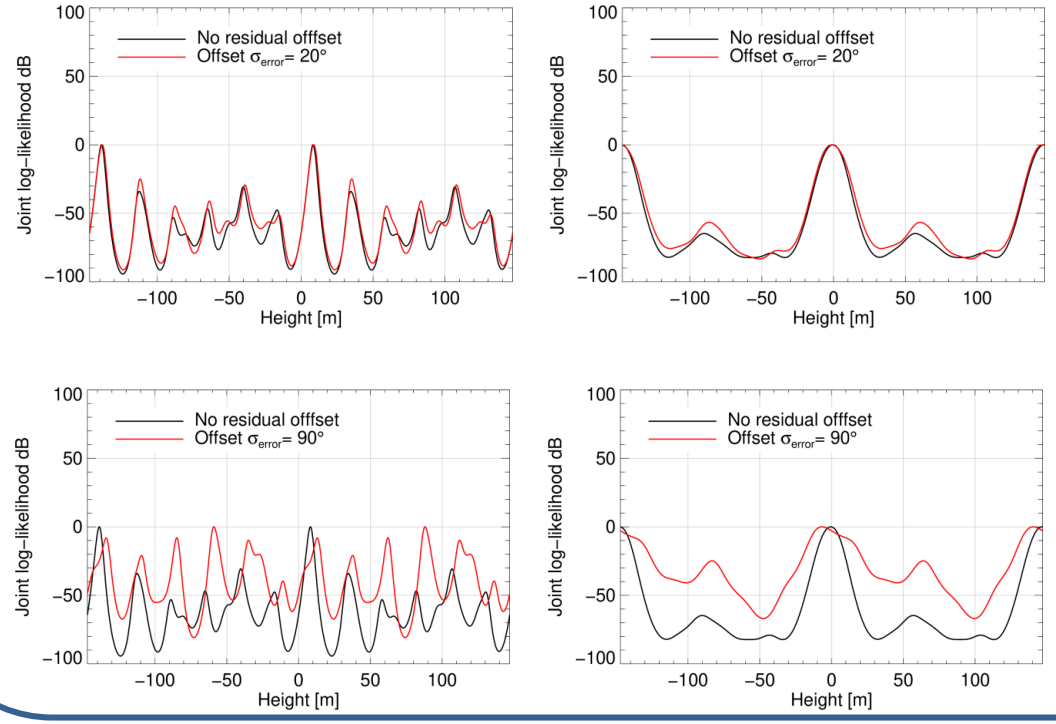


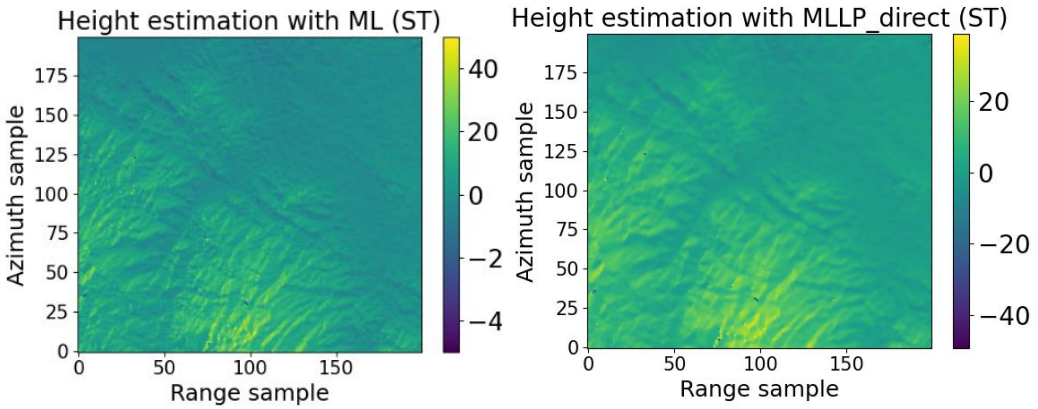
AfriSAR FSAR P-band dataset over Lopè
Large baseline (kz ~ 0.07-0.11; HoA ~ 55-90m)

Height estimation

- Decorrelation due to poor backscatter, geometry or volume (specially for large baselines w.r.t. critical)
- Tackled in three steps: spectral filter, coherence phase optimization & multi-baseline phase center height estimation

Multi-baseline height estimation



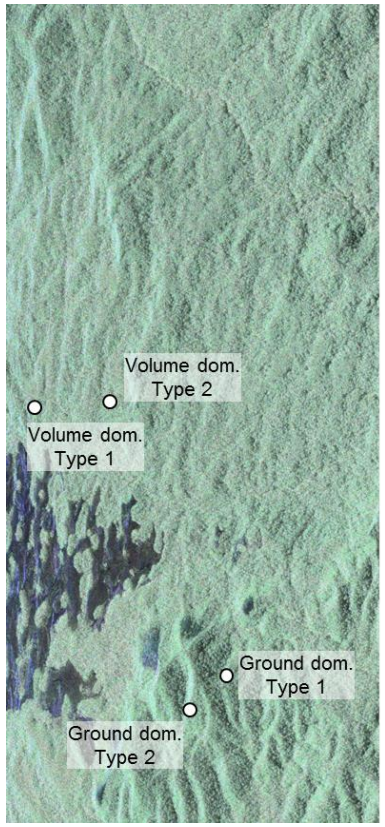


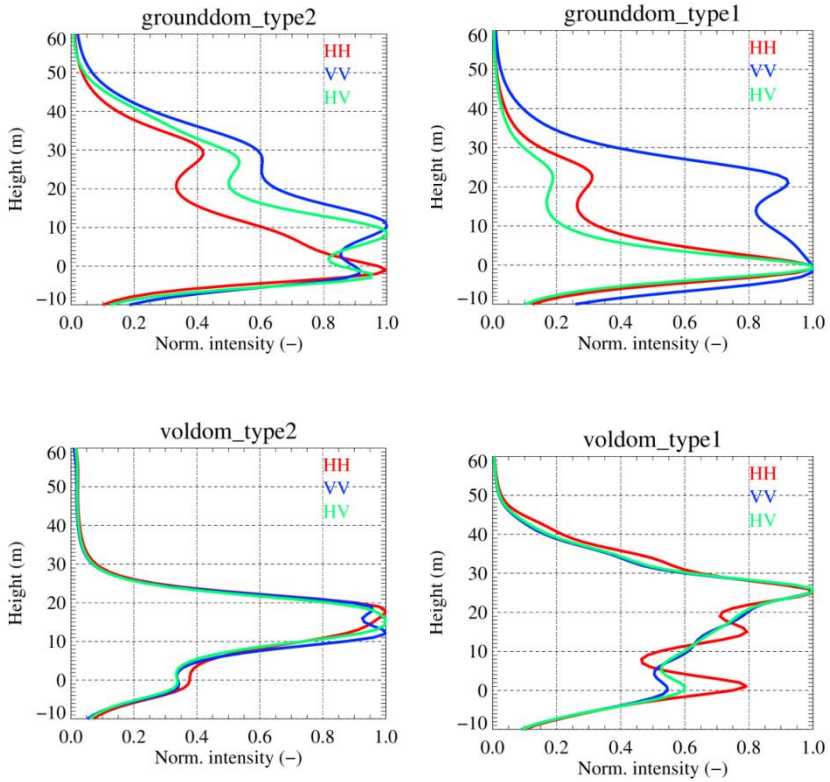
[3] V. Pascazio and G. Schirinzi, "Multifrequency InSAR height reconstruction through maximum likelihood estimation of local planes parameters"

DTM Generation

DEM bias under vegetation

DTM estimation





Phase center DEM estimation with multi-baseline interferometry

↓

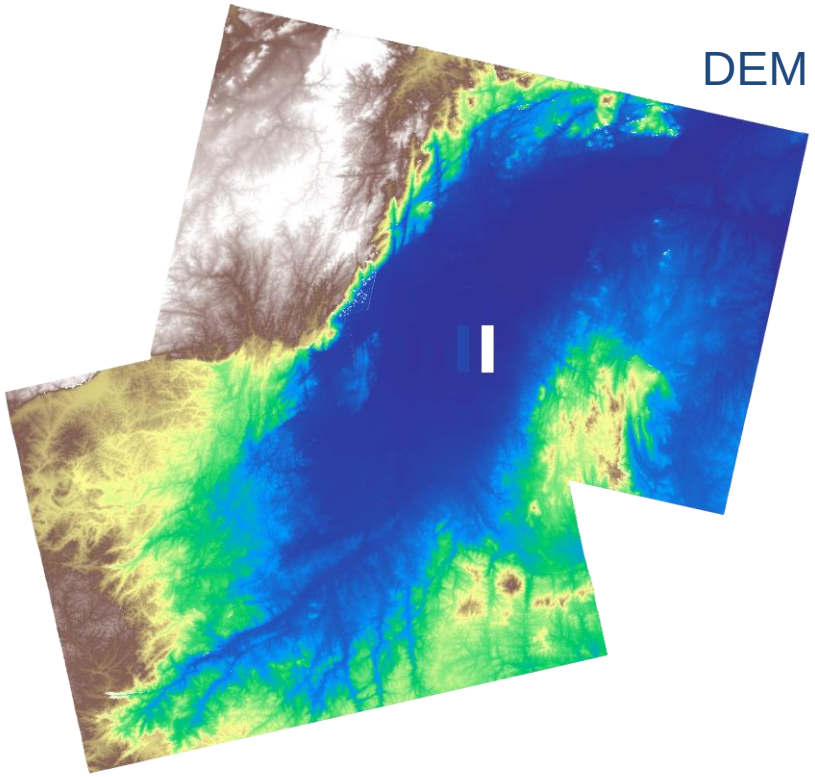
Ground steered phase with Sum of Kronecker Products (SKP)

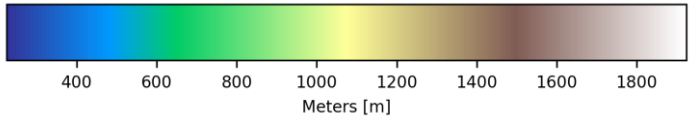
↓

Phase center DTM estimation with multi-baseline interferometry

Mosaicking

Mosaicking


DEM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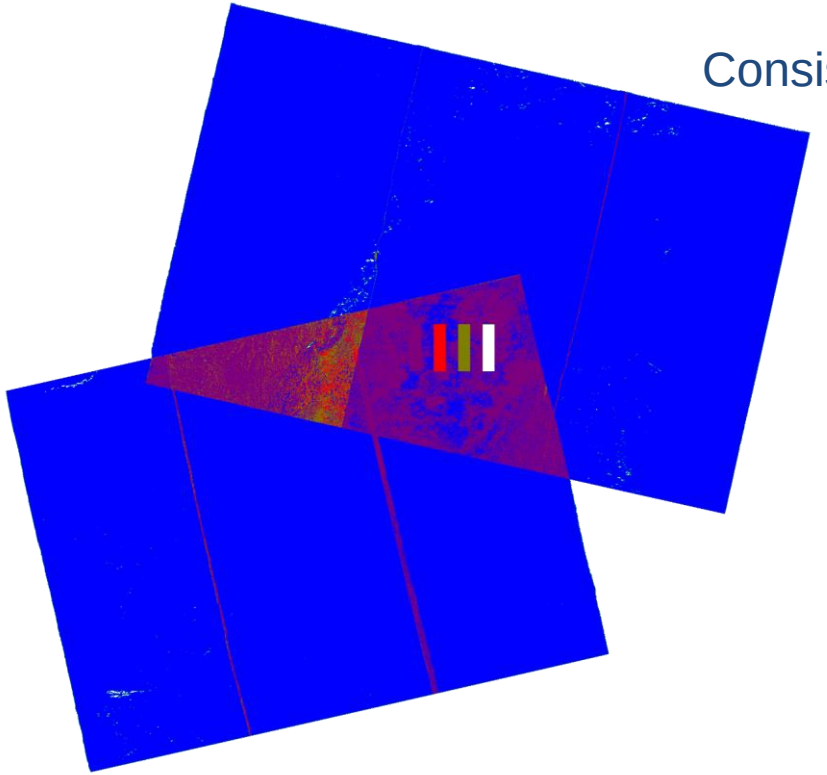
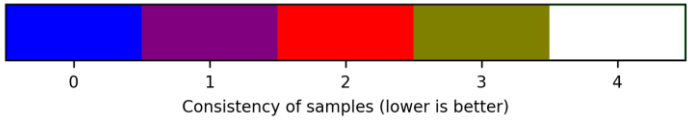
1800
Meters [m]


Consistence

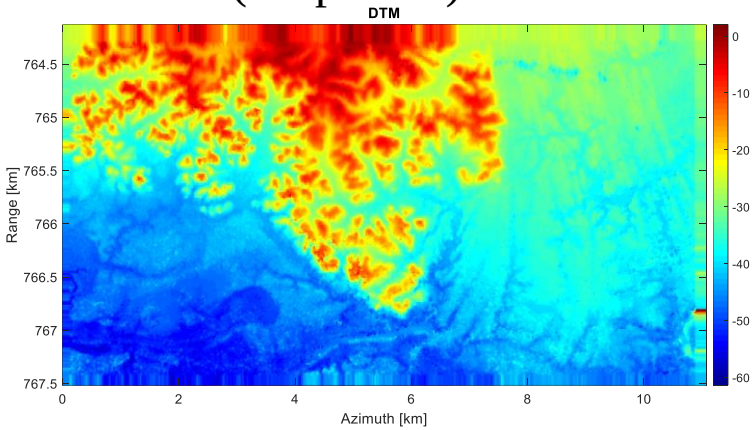

0
1
2
3
4
Consistency of samples (lower is better)

Validation

Validation

Simulation from airborne datasets

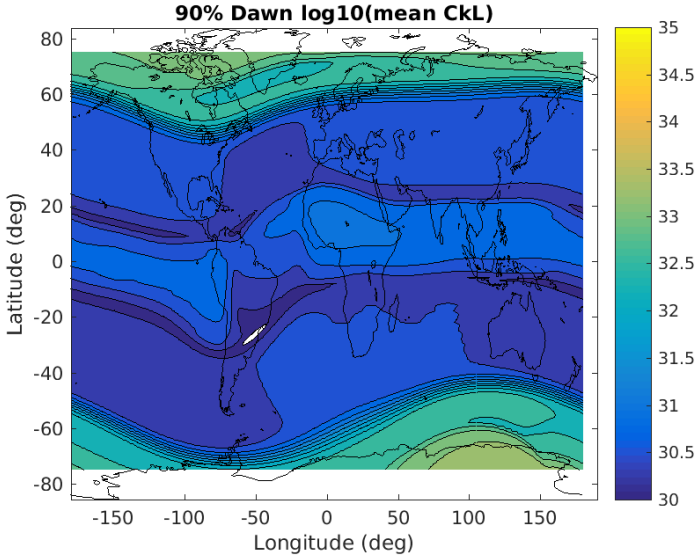
Paracou (TropiSAR)



Location: French Guiana

Forest type: Tropical

Original f_0: $397.5MHz\ (\lambda \approx 75cm)$



No ionosphere

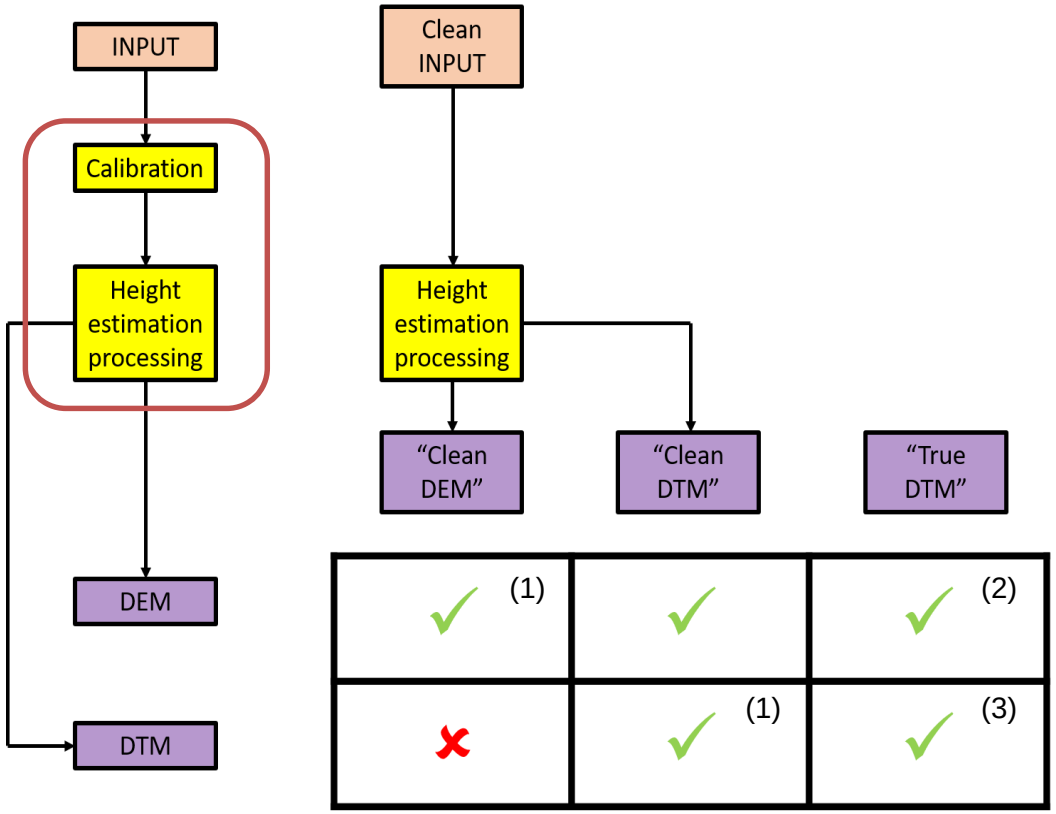
Weak ionosphere

Strong ionosphere

[4] Mariotti d'Alessandro M., Tebaldini S. "Cross Sensor Simulation of Tomographic SAR Stacks", *Remote Sens.* 2019, 11(18), 2019, https://doi.org/10.3390/rs11182099

Validation

Validation strategy

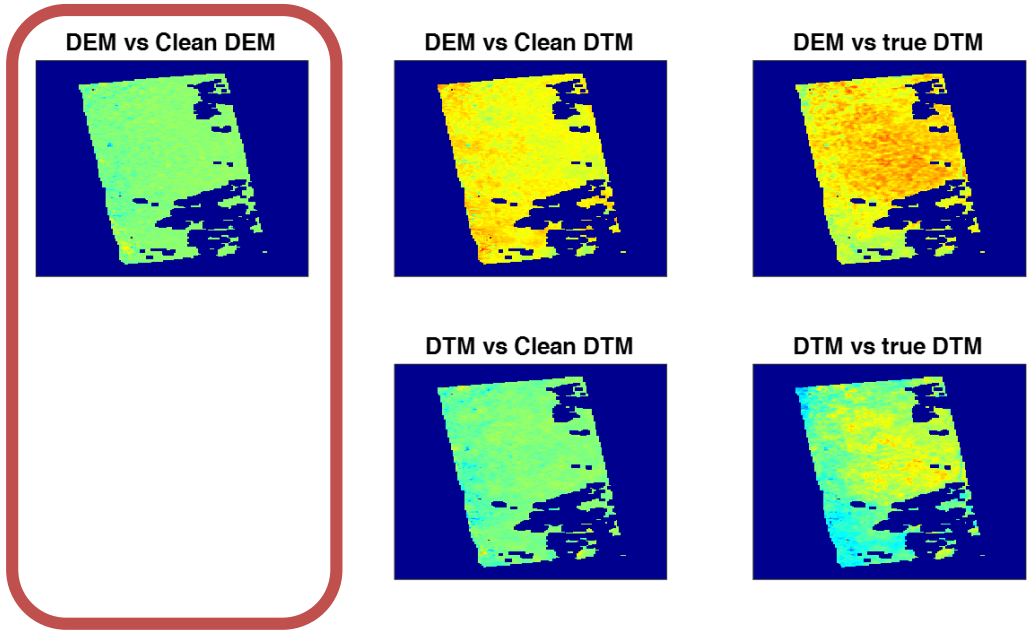


Height differences:

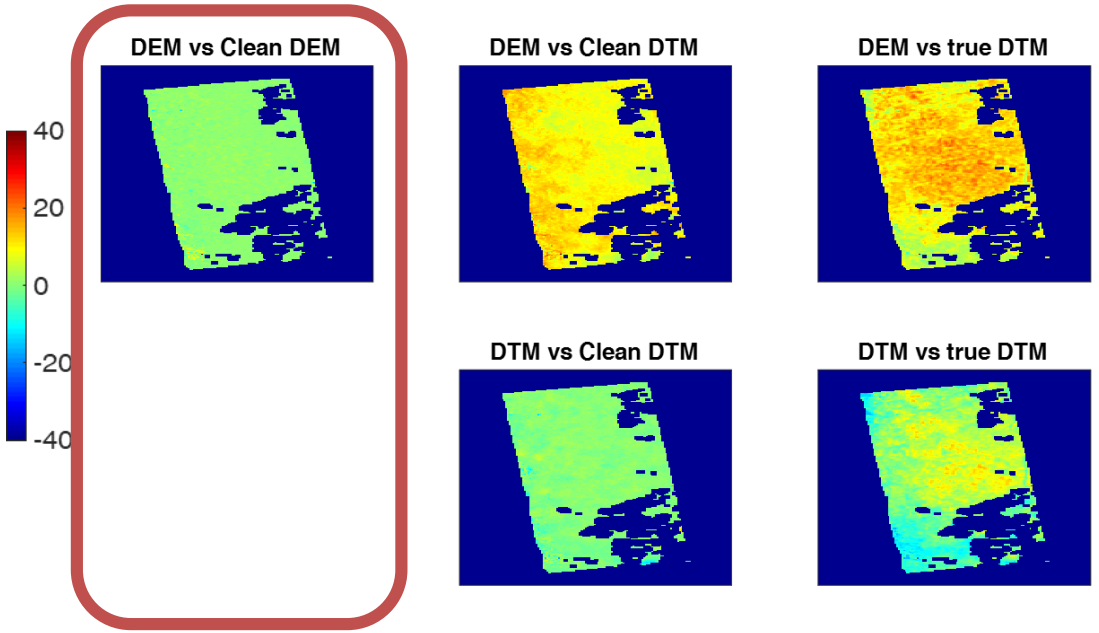
1. Close to zero if the calibration block does not deteriorate the phase center DEM estimates
2. A significant elevation bias is expected as the penetration through the vegetation layer is limited
3. Close to zero if BIODEMPP produces good topographic estimates

Paracou – DEM/DTM differences

Tropical ionosphere scenario



Boreal ionosphere scenario



Summary

- BIOMASS image stacks can be processed to retrieve a **DTM** by exploiting both interferomeric and tomographic approaches
- The BIODEMPP products are:
  - **L3-DTM** (ground topography)
  - **L3-DEM** (phase center height at P band)
  - Coregistered and calibrated TOM/INT stacks as input to the L2 algorithms
- The stacking and calibration modules are being implemented operationally in the frame of the **BPS** (BIOMASS Processing Suite), responsable for the generation of the BIOMASS SAR and biophysical products.
- Still TBD whether the other modules (DEM generation, DTM generation and mosaicking) will be implemented and integrated in the future as part of the BPS.
- The BIODEMPP code shall be made available via **MAAP** (Mission Algorithm and Analysis Platform).

*MAAP: https://earthdata.nasa.gov/esds/maap

Thank you!